Next-Day Rain Classification for the Colombo Area

Machine Learning Assignment - MSc in Artificial Intelligence
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1. Problem definition and dataset

The task is to classify the following day as wet or dry in the Colombo area using recent weather and the time of year. A wet day means at least 1 mm of precipitation. Rain affects transport and outdoor work, but this target does not describe rainfall severity or flood risk.

The dataset was compiled from the Open-Meteo Historical Weather API, selecting ECMWF ERA5 [1]. It contains 4,383 daily records from 1 January 2014 to 31 December 2025, aggregated in Asia/Colombo and downloaded on 27 October 2026. The requested point was 6.9271 N, 79.8612 E; the returned grid location was 7.0 N, 80.0 E. The observations are therefore gridded estimates for the Colombo area, not station measurements.

Feature group	**Description**
Preceding-day rain	Daily precipitation (mm)
Humidity	Daily mean relative humidity (%)
Maximum/minimum temperature	Daily air-temperature extremes (C)
Wind	Daily maximum 10 m wind speed (km/h)
Pressure	Daily mean surface pressure (hPa)
3-day and 7-day mean rain	Trailing means ending on the preceding day (mm/day)
Pressure change	Preceding-day pressure minus the day before (hPa)
Season sine and cosine	Two values calculated from the target date

Preprocessing and ethical use

Dates were unique and consecutive, with no missing raw values. Rainfall was checked for negative values and minimum temperature against maximum temperature. No imputation or outlier removal was needed. The first six incomplete history windows and the final unavailable next-day target were removed, leaving 4,376 rows and 11 numerical predictors. Weather-history features use only the completed day before the target and earlier days. No categorical encoding was needed.

The source is public and contains no personal or sensitive data. Open-Meteo and ECMWF ERA5 are credited under CC BY 4.0 [2]. The source request, units and collection details accompany the dataset.

2. Algorithm selection and training

Regularised Quadratic Discriminant Analysis (QDA) was selected because it is outside the supplied lecture-algorithm list and uses no deep learning or image processing. QDA fits a multivariate Gaussian density to each class, with separate means and covariance matrices, and applies Bayes' rule [3]. Unequal covariances allow a quadratic decision boundary and interactions between weather variables. Regularisation helps when rainfall-history predictors are correlated.

Logistic regression fits a linear log-odds function directly, while QDA models the inputs within each class. A decision tree divides the feature space using rules; QDA uses continuous class densities. KNN compares distances to stored examples, whereas QDA summarises each class through fitted parameters. QDA offers nonlinear classification with a compact model, although its Gaussian assumptions may be unsuitable for skewed rainfall.

Data split and preprocessing

Partition	Target dates	Rows	Wet days
Train	2014-01-08 to 2021-12-31	2915	2116
Validation	2022-01-01 to 2023-12-31	730	572
Test	2024-01-01 to 2025-12-31	731	556

The chronological split avoids mixing neighbouring days across a random train/test split. Target rainfall is excluded from the predictors; only the known target date supplies the season terms. Their angle is 2*pi*(day-of-year - 1)/365.25. The three rainfall predictors receive log(1 + rainfall), then all inputs are standardised using training statistics only. Class priors are training class frequencies; no class resampling is used.

Hyperparameters and evaluation measures

Covariance regularisation alpha was compared at 0, 0.01, 0.05, 0.10, 0.25, 0.50, 0.75 and 1.00 using validation AUC. Each covariance is shrunk towards the identity in standardised coordinates. Alpha 0.75 gave the best validation AUC, 0.8978. Validation balanced accuracy selected a wet-day cutoff of 0.78 from 0.20-0.80 in steps of 0.005. The model was not refitted on validation or test rows.

AUC measures probability ranking. Accuracy is the fraction of correct decisions; balanced accuracy gives wet and dry recall equal weight. F1 combines wet-day precision and recall. Brier loss measures squared probability error and is better when lower. These measures are used together because wet days are common in this dataset.

Comparison baselines used training-month wet frequencies, preceding-day wet/dry persistence and constant training prevalence. Logistic regression used train-only scaling and C = 1. Validation data selected the logistic and monthly-climatology cutoffs.

3. Evaluation results

Model	AUC	Accuracy	Balanced accuracy	F1	Brier loss
Regularised QDA	0.873	0.814	0.786	0.873	0.132
Logistic regression	0.885	0.807	0.801	0.865	0.107
Monthly climatology	0.825	0.811	0.807	0.868	0.131
Rain persistence	0.775	0.836	0.775	0.892	0.164
Constant train prevalence	0.500	0.761	0.500	0.864	0.183

The test set contains 731 days: 556 wet and 175 dry. QDA achieved AUC 0.873 and F1 0.873. Logistic regression gave higher AUC and lower Brier loss. Persistence gave higher accuracy and F1, while monthly climatology gave higher balanced accuracy. QDA therefore did not outperform all simpler alternatives.

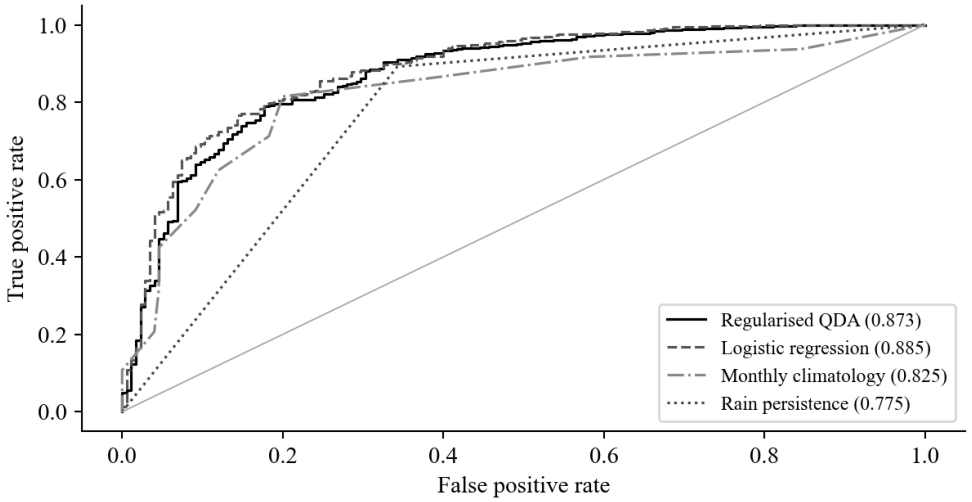


Figure 1. ROC curves for the 2024-2025 test period. The diagonal shows chance ranking.

Confusion matrix at the 0.78 cutoff

Actual / predicted	Dry	Wet
Dry	128	47
Wet	89	467

There were 467 correct wet predictions, 128 correct dry predictions, 89 missed wet days and 47 false wet alerts. Wet-day recall was 83.99%, dry-day recall 73.14% and wet-day precision 90.86%. Predicting wet every time would give 76.06% accuracy, which explains the need for the additional measures.

4. Explainability and interpretation

Interventional SHAP was used to explain QDA's raw log-odds score [4]. The background consists of all 2,915 training rows. Since the score is quadratic, linear, squared and pairwise terms can be allocated exactly using the Shapley rule. The base value and feature contributions reconstruct the raw score; a logistic transform produces the probability. Contributions do not add directly to the displayed percentage.

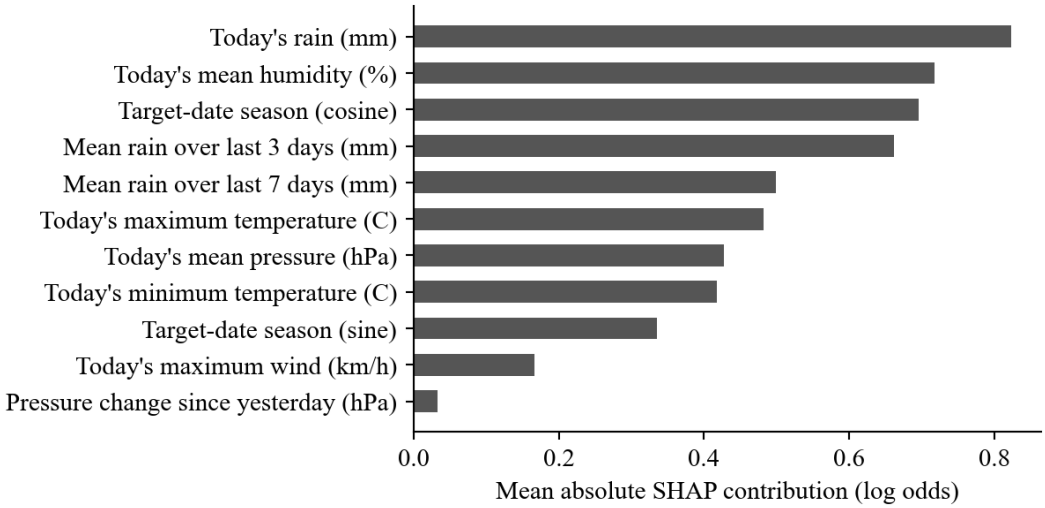


Figure 2. Mean absolute SHAP contributions for 240 evenly spaced test days.

Preceding-day rain had the largest mean absolute contribution (0.823 log odds), followed by humidity (0.718) and cosine season (0.696). Recent rain, moisture and annual seasonality are plausible influences on rain-day classification. This agrees with domain knowledge at the level of associations; it does not establish a causal weather mechanism.

Individual predictions

Example	Target date	p(wet)	Actual rain
Correct wet day	2025-01-01	90.5%	9.4 mm
False wet alert	2025-01-20	99.9%	0.7 mm

For 1 January 2025, preceding-day rain was 6.6 mm and humidity was 85%. Humidity contributed +0.568 log odds and the cosine season term contributed -1.020. The base score 2.273 plus all contributions gave 2.257, corresponding to a 90.5% wet-day probability. The recorded 9.4 mm made this a correct wet prediction.

The false alert on 20 January 2025 received 99.9% probability, although only 0.7 mm fell. An explanation describes the fitted model's behaviour, not whether it is correct. Related rain windows and the two season coordinates can share credit, and interventional replacements may create unrealistic combinations.

5. Critical discussion

Data availability is the main limitation. ERA5 is retrospective reanalysis, so complete inputs would not necessarily have been available when a next-day forecast was issued. The chronological split prevents target leakage but does not establish operational forecasting skill. The returned grid cell also differs from the requested point and cannot represent every local shower. Station measurements would be needed to assess local accuracy.

QDA assumes Gaussian class densities. Rainfall has many small values and a long upper tail, which log transformation cannot fully correct. Rolling rain windows overlap and seasonal coordinates are dependent. Covariance regularisation reduces instability but does not make these assumptions true. The stronger logistic-regression AUC and Brier result show that extra model flexibility did not automatically improve generalisation.

Wet days dominate the data, so aggregate accuracy can hide weaker dry-day performance. A single-city model also gives geographically uneven coverage. The 1 mm boundary separates small amounts such as 0.7 mm and 1.7 mm into different classes. Only two years were held out for testing, and validation rows were reused for hyperparameter and cutoff selection. These choices limit confidence in wider use.

Probabilities were not independently calibrated. Confident wrong outputs could mislead people planning outdoor work, and false wet alerts and missed rain have different costs. The application is therefore presented as a historical research demonstration, with no flood or safety advice. Calibration and checks with time-available local inputs would be necessary before practical use.

6. Frontend integration

The Streamlit frontend accepts nine weather-history inputs and a target date, which supplies two season values. It displays a wet/dry classification, model probability and SHAP explanation for the actual entry. Users can select a recorded example or edit the inputs and submit again. Numerical limits, temperature consistency and rolling rain totals are checked before prediction. The interface also shows the evaluation results and dataset source.

References

[1] Open-Meteo. Historical Weather API documentation.

[2] Creative Commons. Attribution 4.0 International licence.

[3] Scikit-learn. QuadraticDiscriminantAnalysis, version 1.8 documentation.

[4] Lundberg, S. M. and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems, 30.